

$$\begin{array}{ccccccc}
\mathcal{H}(t) < \infty & \Leftrightarrow & \begin{array}{c} \Upsilon(\beta) < \infty \\ \text{Dalang's} \\ \text{condition (12)} \end{array} & \Leftarrow & \begin{array}{c} \Upsilon_\alpha(\beta) < \infty \\ \text{Condition (13)} \end{array} & \Leftrightarrow & \begin{array}{c} \mathcal{H}_{\alpha/2}(t) < \infty \\ \text{Condition (31)} \end{array} \\
\Uparrow & & \Uparrow & & \Uparrow & & \Uparrow \\
\lim_{t \rightarrow \infty} \mathcal{H}(t) < \infty & \Leftrightarrow & \begin{array}{c} \Upsilon(0) < \infty \\ \text{Condition (10a)} \end{array} & & \begin{array}{c} \Upsilon_\alpha(0) < \infty \\ \text{Condition (30)} \end{array} & \Leftrightarrow & \lim_{t \rightarrow \infty} \mathcal{H}_{\alpha/2}(t) < \infty
\end{array}$$